

**Academic Stress and Mental Health Among Urban Chinese Adolescents:
An Analysis Based on the Biopsychosocial Model**

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Adolescent mental health has become a major issue in the global public health field. In China, urban adolescents are under a unique academic pressure rooted in the social and cultural structure (Kieling et al., 2011; Sun, Dunne & Hou, 2013). Unlike Western adolescents, who more often face issues such as peer relationships or identity, the pressure on Chinese adolescents mainly comes from academic performance. This pressure starts to appear early and reaches to peak as they approach the Gaokao, the exam that largely determines individual's future opportunities.

Given China's large population base, competition is becoming increasingly fierce, which in turn has led to higher academic requirements. Chinese adolescents are experiencing an increase in the prevalence of anxiety and depression (Marani et al., 2020). According to Sun Hongyan (2024) in *People's Tribune*, two major factors influencing young people's mental health today are employment and academic pursuits. It can also be found in the *China National Mental Health Development Report (2021-2022)* that the risk of adolescents suffering from depression is even higher than that of adults, and many college students report feeling stressed due to academic transitions and future uncertainties.

This article uses the **biopsychosocial model** proposed by Engel (1977) to explain these relationships. Engel argued that health and illness cannot be just attributed itself to biological factors; psychological states and social environments are equally important. The framework is applicable here because the academic stress of urban Chinese adolescents is not just a biological or psychological phenomenon, but is shaped by family expectations, school

policies, and cultural values. **Under the biopsychosocial model, this article explores how academic stress affects the mental health of urban Chinese adolescents from three interrelated dimensions: the biological dimension, the psychological dimension, and the social dimension.**

Background

The academic pressure in urban areas of China is deeply affected by the social structure. In modern China, the college entrance examination (National College Entrance Examination) has always been the official channel for accessing higher education and even achieving social mobility. With families making significant financial investments in their children's education, it has become commonplace for kids to attend after-school private tutoring, advanced training, and exam-focused courses (Liu, 2015). This pressure is further increased by schools' frequent exam schedules, public announcement of grade rankings, and curriculum orientation that prioritizes exam preparation.

The information gleaned from comparison is stronger. Nearly half of teenagers with depressive symptoms reported frequently wanting to leave the school setting, compared to less than 5% of teenagers without depressive symptoms, according to a study conducted by the Institute of Psychology, Chinese Academy of Sciences (2025). We can draw the conclusion that **academic stress and mental health are positively correlated**. A study headed by Xu (2023) came to the same conclusion and was published in the journal *Developmental Psychology*. In a long-term follow-up of Chinese teenagers, the team discovered that academic stress causes adolescents' sleep quality to deteriorate and increases their susceptibility to negative emotions. Even worse, these effects will compound over time.

George Engel (1977) proposed the biopsychosocial model, which further revised the biomedical approach that neglects the psychological and social aspects of diseases. This model views health as a result of the interaction between biological factors psychological factors, and social factors. In research on child and adolescent mental health, this model has unique value in understanding how multiple risks overlap and reinforce each other (Borrell-Carrió, Suchman, & Epstein, 2004).

Biological Dimension

When teenagers face long-term academic pressure, their bodies will have obvious adverse reactions. At the biological level, this is because academic stress disrupts normal neurochemistry, particularly through the overactivation of the HPA axis, leading to elevated cortisol levels. Stroud and colleagues (2009) found that students facing big exams had great elevated cortisol levels, which in turn harm their attention and memory. Over time, this pattern can evolve into anxiety and depression.

The brain itself also changes under continuous stress. Studies using functional magnetic resonance imaging have shown that prolonged stress overactivates the amygdala, which is in charge of processing emotions like anxiety and fear. Furthermore, it inhibits the activity of the prefrontal cortex, which helps regulate these emotions (Lupien et al., 2009). As a result, the brain's capacity for self-regulation is weakened and it reacts to stress more strongly. For teenagers who are already under stress, that is a dangerous combination.

The impact of stress on the body also happens in more obvious ways. Sleep is often the first to be affected. According to researchers like Yan (2018), academic pressure is the primary cause of sleep quality issues, which affect about 25% of middle school students.

Students who don't get enough sleep are more likely to experience physical discomforts like headaches, stomachaches, and fatigue, which exacerbates their physical symptoms and increases their psychological burden. This creates a vicious cycle where stress interferes with sleep, sleep deprivation damages health, and declining health makes it more difficult for people to handle stress.

Psychological Dimension

Beyond the biological impacts described above, academic achievement shapes the self-worth of many Chinese urban teenagers beyond test scores. Success means being regarded as a "good student" or a "good child," while failure may lead them to feel that their self-worth is completely diminished. Zhang and Csizmadia (2024) refer to this as an academic self-worth orientation, in which self-identity is tightly tied to academic performance. When students fail to meet their own or others' expectations, they often experience intense negative emotions such as guilt, powerlessness, and self-criticism. This pattern reflects a maladaptive belief system that links self-esteem entirely to academic success.

Some students further develop maladaptive perfectionism. They set unrealistically high standards, are overly sensitive to minor failures, and constantly compare themselves with others (Flett & Hewitt, 2022). In the highly competitive environment of urban schools in China, such traits are often praised as signs of diligence and ambition, even though they increase anxiety. Maladaptive perfectionism is especially likely to emerge among students with a temperament oriented toward high standards.

Adolescence is already a period characterized by strong emotional swings, which are closely related to biological changes. Chronic stress hinders the development of healthy

coping strategies. As Compas (2017) demonstrated in a systematic meta-analysis, stressed adolescents are more likely to engage in rumination about unpleasant memories instead of focusing on problem-solving. When individuals ruminate, they become less aware of the need to seek help or solutions, trapping themselves in a cycle of worry and avoidance. Once this pattern is established, it may persist into adulthood and beyond.

Social Dimension

While these psychological responses take shape at the individual level, adolescent stress and maladaptive coping patterns do not exist in isolation. For most urban Chinese adolescents, stress begins from home. Parents not only invest money in their children's education but also pour in emotional energy, and the resulting expectations are often suffocating. In family conversations, the topic of grades has replaced other communication content, and their leisure time is always filled with cram schools and homework (Marani et al., 2020). Even well-meaning parents may not realize that their over-involvement for example: "If you finish your homework early, you can rest early." is exacerbating their children's stress. At this time, if the child resists, they will violate both diligence in study and filial piety (Zhang and Csizmadia, 2024).

Schools also play an important role. The ranking system deliberately reminds students to always pay attention to their position among their peers and sends a signal: only students with top grades are worthy of attention and affection. Sun, Deng, and Hou (2013) pointed out that such an environment will damage psychological security, leaving students in a state of constant external evaluation and doubt about their own value.

Behind all this lies China's traditional culture. Confucianism's emphasis on education has deep historical roots, and the imperial examination system has taken on new forms in contemporary China. Since ancient times, people have generally believed that academic qualifications determine social mobility, and contemporary social media has further reinforced this concept by showcasing "all-round excellent students" and "success stories from prestigious schools" (Zhang and Csizmadia, 2024). Adolescents receive and internalize this information, gradually viewing pressure as a normal or even necessary part of life.

Conclusion

This article examines the relationship between academic stress and the mental health of Chinese urban adolescents using Engel's biopsychosocial model. At the biological level, it primarily shows up as physiological symptoms like sleep disturbances, hormonal imbalances, and changes in brain function. At the psychological level, it manifests as students' inability to control their emotions and their tendency to link their self-worth to their grades. Some even develop a perfectionism pattern. Achievement-oriented family education, a competitive school system, and a culture that links success to academic credentials are just a few examples of the widespread social effects. These levels do not exist in isolation; they interact with and reinforce each other, forming a dynamically interconnected network of influences.

The uniqueness of this case lies in its cultural specificity. The pressures that Chinese urban adolescents face might not be considered a universal phenomenon, but rather a reflection of the unique influences of Chinese education and society. At the same time, by demonstrating the interplay between biological, psychological, and social factors, the bio-psycho-social model has further validated its relevance today. This also serves as a reminder

that by comprehending adolescent mental health issues, we can examine them from these three perspectives, regardless of where we live in China.

References

- Borrell-Carrió, F., Suchman, A. L., & Epstein, R. M. (2004). The biopsychosocial model 25 years later: Principles, practice, and scientific inquiry. *Annals of Family Medicine*, *2*(6), 576–582. <https://doi.org/10.1370/afm.245>
- Compas, B. E., Jaser, S. S., Bettis, A. H., Watson, K. H., Gruhn, M. A., Dunbar, J. P., Williams, E., & Thigpen, J. C. (2017). Coping, emotion regulation, and psychopathology in childhood and adolescence: A meta-analysis and narrative review. *Psychological Bulletin*, *143*(9), 939–991. <https://doi.org/10.1037/bul0000110>
- Engel, G. L. (1977). The need for a new medical model: A challenge for biomedicine. *Science*, *196*(4286), 129–136. <http://www.jstor.org/stable/1743658>
- Flett, G. L., & Hewitt, P. L. (2022). Introduction: Perfectionism in childhood and adolescence. In G. L. Flett & P. L. Hewitt (Eds.), *Perfectionism in childhood and adolescence: A developmental approach* (pp. 3–8). American Psychological Association. <https://doi.org/10.1037/0000289-001>
- Institute of Psychology, Chinese Academy of Sciences. (2025). *2024 Report on adolescent mental health and academic status* [in Chinese]. <https://psych.cas.cn/>
- Kieling, C., Baker-Henningham, H., Belfer, M., Conti, G., Ertem, I., Omigbodun, O., ... & Rahman, A. (2011). Child and adolescent mental health worldwide: Evidence for

- action. *The Lancet*, *378*(9801), 1515–1525. [https://doi.org/10.1016/S0140-6736\(11\)60827-1](https://doi.org/10.1016/S0140-6736(11)60827-1)
- Liu, X. (2015). Who has more opportunities to attend college?—An empirical study of the strata distribution of different qualities of higher education opportunities in China. *Chinese Education & Society*, *48*(3), 201–217. <https://doi.org/10.1080/10611932.2015.1085769>
- Lupien, S. J., McEwen, B. S., Gunnar, M. R., & Heim, C. (2009). Effects of stress throughout the lifespan on the brain, behaviour and cognition. *Nature Reviews Neuroscience*, *10*(6), 434–445. <https://doi.org/10.1038/nrn2639>
- Marani, H., Fujioka, J., Tabatabavakili, S., & Bollegala, N. (2020). Systematic narrative review of pediatric-to-adult care transition models for youth with pediatric-onset chronic conditions. *Children and Youth Services Review*, *118*, 105415. <https://doi.org/10.1016/j.childyouth.2020.105415>
- Sun, H. (2024, April 20). Analysis of factors affecting adolescent mental health and countermeasures [in Chinese]. *People's Tribune*. https://paper.people.com.cn/rmlt/html/2024-04/20/content_26070945.htm
- Sun, J., Dunne, M., Hou, X., & Xu, A. (2013). Educational stress among Chinese adolescents: Individual, family, school and peer influences. *Educational Review*, *65*(3), 284–302. <https://doi.org/10.1080/00131911.2012.659657>

- Stroud, L. R., Foster, E., Papandonatos, G. D., Handwerger, K., Granger, D. A., Kivlighan, K. T., & Niaura, R. (2009). Stress response and the adolescent transition: Performance versus peer rejection stressors. *Development and Psychopathology*, *21*(1), 47–68. <https://doi.org/10.1017/S0954579409000042>
- Xu, J., Wang, H., Liu, S., Hale, M. E., Weng, X., Ahemaitijiang, N., Hu, Y., Suveg, C., & Han, Z. R. (2023). Relations among family, peer, and academic stress and adjustment in Chinese adolescents. *Developmental Psychology*, *59*(7), 1346–1358. <https://doi.org/10.1037/dev0001538>
- Yan, Y. W., Lin, R. M., Su, Y. K., & Liu, M. Y. (2018). The relationship between adolescent academic stress and sleep quality: A multiple mediation model. *Social Behavior and Personality: An International Journal*, *46*(1), 63–78. <https://doi.org/10.2224/sbp.6530>
- Zhang, F., & Csizmadia, A. (2024). Academic socialization model: Understanding Chinese children's academic self-concept and the role of academic achievement. *Journal of Family Theory & Review*, *16*(4), 684–694. <https://doi.org/10.1111/jftr.12584>
- 中国科学院心理研究所. (2022). 心理健康蓝皮书（2022 版）发布：青少年群体抑郁风险高于成年群体 [Report on national mental health development in China (2021-2022)]. https://psych.cas.cn/news/cmsm/202303/t20230301_6687013.html